Fermi–Amaldi-Based Local Mixing Function

Ivan P. Bosko and Viktor N. Staroverov
Department of Chemistry, The University of Western Ontario, London, Ontario N6A 5B7, Canada
(Dated: March 6, 2026)

I. MAIN TEXT

We propose a new local mixing function (LMF) for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities:

$$\mathrm{LMF} = \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}}. \tag{1}$$

The (closed-shell) exact exchange energy density in terms of a one-electron reduced density matrix, $\gamma(\mathbf{r};\mathbf{r}') = 2\sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}')$, is given by [1, 2]

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = -\frac{1}{4}\int \frac{|\gamma(\mathbf{r};\mathbf{r}')|^2}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}'. \tag{2}$$

The Fermi–Amaldi [3, 4] energy density is expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N,

$$e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{1}{2N}\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}'. \tag{3}$$

Both exchange densities are manifestly non-positive and equivalent for one-electron systems and two-electron singlets. Although, for $N > 2$, one can show that

$$e_{\mathrm{x}}^{\mathrm{exact}} \leq e_{\mathrm{x}}^{\mathrm{FA}} \leq 0, \tag{4}$$

since $e_{\mathrm{x}}^{\mathrm{FA}}$ is scaled down by N. Combining these properties, we find that the proposed LMF is bound between zero and one:

$$0 \leq \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} \leq 1. \tag{5}$$

This inequality highlights that for one-electron systems the LMF is unity, meaning one hundred percent exact exchange. A particular case of a one-electron limit, i.e., the asymptotic limit ($r \to \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r\to\infty} \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} = 1 \tag{6}$$

since both energy densities approach $-1/2r$ as $r \to \infty$.

Another important constraint, the high-density limit,

[1] V. N. Staroverov, Density-functional approximations for exchange and correlation, in *A Matter of Density* (John Wiley & Sons, Ltd, 2012) Chap. 6, pp. 125–156.
[2] R. Parr and Y. Weitao, *Density-Functional Theory of Atoms and Molecules*, International Series of Monographs on Chemistry (Oxford University Press, 1989).
[3] E. Fermi and E. Amaldi, Le orbite ∞s degli elementi, Mem. Reale Accad. Italia **6**, 117 (1934).
[4] P. Bénilan, J. A. Goldstein, and G. R. Rieder, The fermi-amaldi correction in spin polarized thomas-fermi theory, in *Differential Equations and Mathematical Physics*, Mathematics in Science and Engineering, Vol. 186 (Elsevier, 1992) pp. 25–37.